

Section 2 — Exploring single-cell RNA-seq of immune cells

across species

Curating species for cross-species expression of paralogous TFs

Paralogous-TF project (iteration 11, computational pipeline)

2026-06-16

1 Question

One goal of this project is to evaluate the expression of paralogous transcription factors (TFs) in immune cells using available bulk or single-cell RNA data, and to do so *beyond human and mouse*. This note inventories which other species have single-cell RNA-seq (scRNA-seq) of immune cells (blood, spleen, lymph node, bone marrow / thymus), starting from the EBI Single Cell Expression Atlas (SCEA) and then widening to all public resources, and places those species on a time-calibrated evolutionary tree so the phylogenetic coverage — and its gaps — is explicit.

2 What is in the EBI Single Cell Expression Atlas

SCEA (the single-cell arm of EMBL-EBI's Expression Atlas) uniformly re-processes every dataset through one pipeline (SMART-like → Kallisto; droplet 10x/ Drop-seq → Alevin/Salmon + **emptyDrops**; downstream SCANPY for clustering and markers), so matrices are comparable across studies and species. A live pull of the experiment list (www.ebi.ac.uk/gxa/sc/json/experiments, saved to `inputs/scea_experiments.json`) returned **383 experiments across 22 species**. Of these, **9 are vertebrates**, and only **5 carry a genuine immune-cell dataset** (Table 1); the other four vertebrates in SCEA (marmoset, sheep, chicken, *Xenopus*) contain only non-immune tissues (embryo, gut epithelium, limb, spinal cord).

Within SCEA the immune comparison is therefore effectively **human/mouse-centric**, with *zebrafish* as the single deep outgroup (kidney marrow = the functional analogue of mammalian bone marrow, plus spleen and blood) and rat/rabbit as thin, single-study additions (bone-marrow mononuclear phagocytes). Figure 1 shows these on the vertebrate tree.

3 The broader cross-resource landscape

The species that matter most for a cross-species paralog comparison — non-human primates and large mammals with secondary-lymphoid-organ coverage — are *not* in SCEA; they live in dedicated atlases and papers. Widening to all public resources adds 14 species (Table 2, Figure 2), giving **19 vertebrate species spanning ~429 Myr of evolution**. The strongest non-human/mouse options with true spleen/lymph-node/marrow coverage are **rhesus macaque** (RIRA), **pig** (porcine lymphoid-organ atlas), **cattle** (Cattle Cell Atlas), and **mouse lemur** (Tabula Microcebus). Most

Table 1: Vertebrate species in SCEA and their immune-cell scRNA-seq content. Immune experiments were counted by matching dataset descriptions against an immune / haematopoietic keyword set.

Common name	Species	Immune exps.	Total exps.
Human	<i>Homo sapiens</i>	44	159
Mouse	<i>Mus musculus</i>	24	126
Zebrafish	<i>Danio rerio</i>	6	15
Rat	<i>Rattus norvegicus</i>	2	3
Rabbit	<i>Oryctolagus cuniculus</i>	1	2
Marmoset	<i>Callithrix jacchus</i>	0	1
Sheep	<i>Ovis aries</i>	0	1
Chicken	<i>Gallus gallus</i>	0	4
Frog (Xenopus)	<i>Xenopus tropicalis</i>	0	1

remaining phylogenetic breadth (carnivores, ungulates, birds) exists only as **blood/PBMC** snapshots from cross-species PBMC and livestock studies — adequate for circulating-TF expression, thin for tissue-resident immune programs.

4 How to obtain and process the data

SCEA is fully open (no login). For each experiment, the per-experiment FTP folder https://ftp.ebi.ac.uk/pub/databases/microarray/data/atlas/sc_experiments/<ACCESSION>/

provides MatrixMarket matrices (*.aggregated_filtered_normalised_counts.mtx + .mtx_rows gene IDs / .mtx_cols barcodes), per-cell metadata (*.cell_metadata.tsv), clustering (*.clusters.tsv), per-cluster marker genes (*.marker_genes_⟨K⟩.tsv), MAGE-TAB sample metadata (*.sdrf.txt), embeddings, and a ready-made *.project.h5ad (AnnData) loadable directly with SCANPY. Matrices are keyed by species-specific Ensembl gene IDs, so the cross-species workflow is: pick experiments per species, map ETS/paralog orthologs to each species’ Ensembl IDs, then summarise the chosen rows by the supplied cell-type / cluster labels. The uniform processing means paralog expression patterns can be compared without re-running quantification. External atlases (RIRA, pig, cattle, Tabula Microcebus) distribute their own h5ad/Seurat objects via GEO/ArrayExpress/portals.

5 Implications for the project

- For **tissue-resident** immune expression of paralogous TFs, the usable species beyond human/mouse are macaque, pig, cattle, mouse lemur, and zebrafish (deep outgroup) — all with spleen/lymph-node/marrow coverage and tractable ortholog mapping.
- The **primate axis** is well sampled (mouse lemur → macaque → human), which is convenient because ETS-family paralog orthology is cleanest within primates.
- Everything else (carnivores, most ungulates, birds) is **blood/PBMC-only** — fine for circulating-cell TF expression, weak for resident programs.

Methods and provenance

SCEA inventory: scripts/05_scea Vertebrate Immune Tree.py reads inputs/scea_experiments.json (snapshot 2026-06-16), counts immune experiments per species by keyword, and renders Figure 1 with the table results/intermediate/scea_Vertebrate Immune Experiments.tsv (includes the exact immune experiment accessions). Cross-resource view: scripts/06_cross_resource Immune Tree.py renders Figure 2 and results/intermediate/cross_resource Immune Species.tsv. Tree topol-

Vertebrates with single-cell RNA-seq of immune cells in the EBI Single Cell Expression Atlas

5 of 9 SCEA vertebrate species carry an immune-cell dataset | filled bubble = # immune scRNA-seq experiments; open grey = in SCEA but no immune dataset

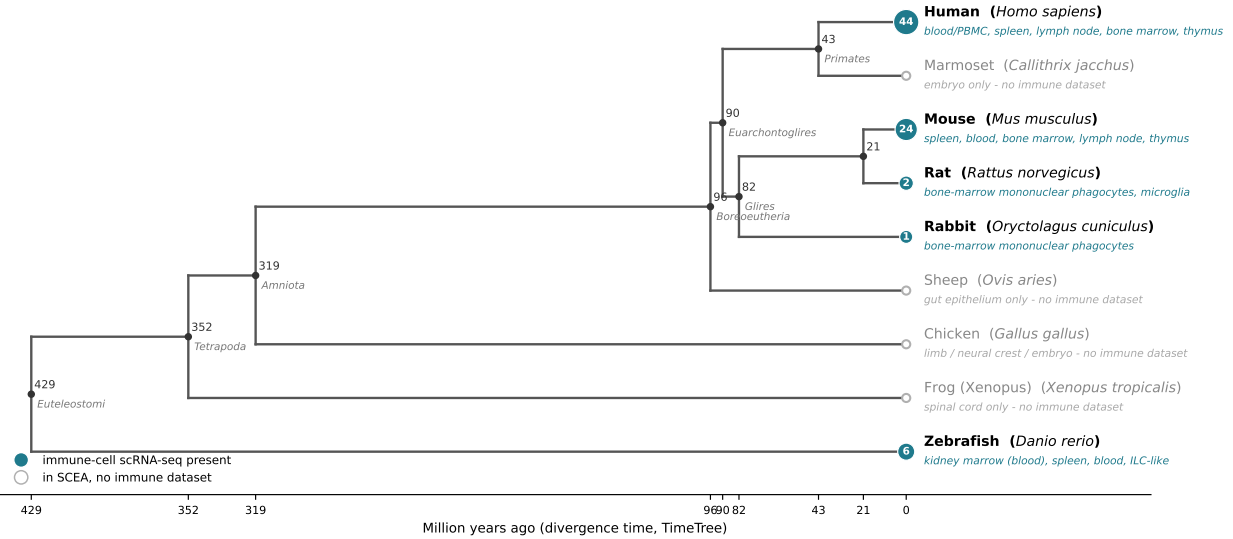


Figure 1: Vertebrates with immune-cell scRNA-seq *in SCEA*, on a time-calibrated species tree (present at right). Filled teal bubbles mark species with immune data (bubble label = number of immune scRNA-seq experiments); open grey tips are in SCEA but have no immune dataset. Divergence times (Mya) from TimeTree.

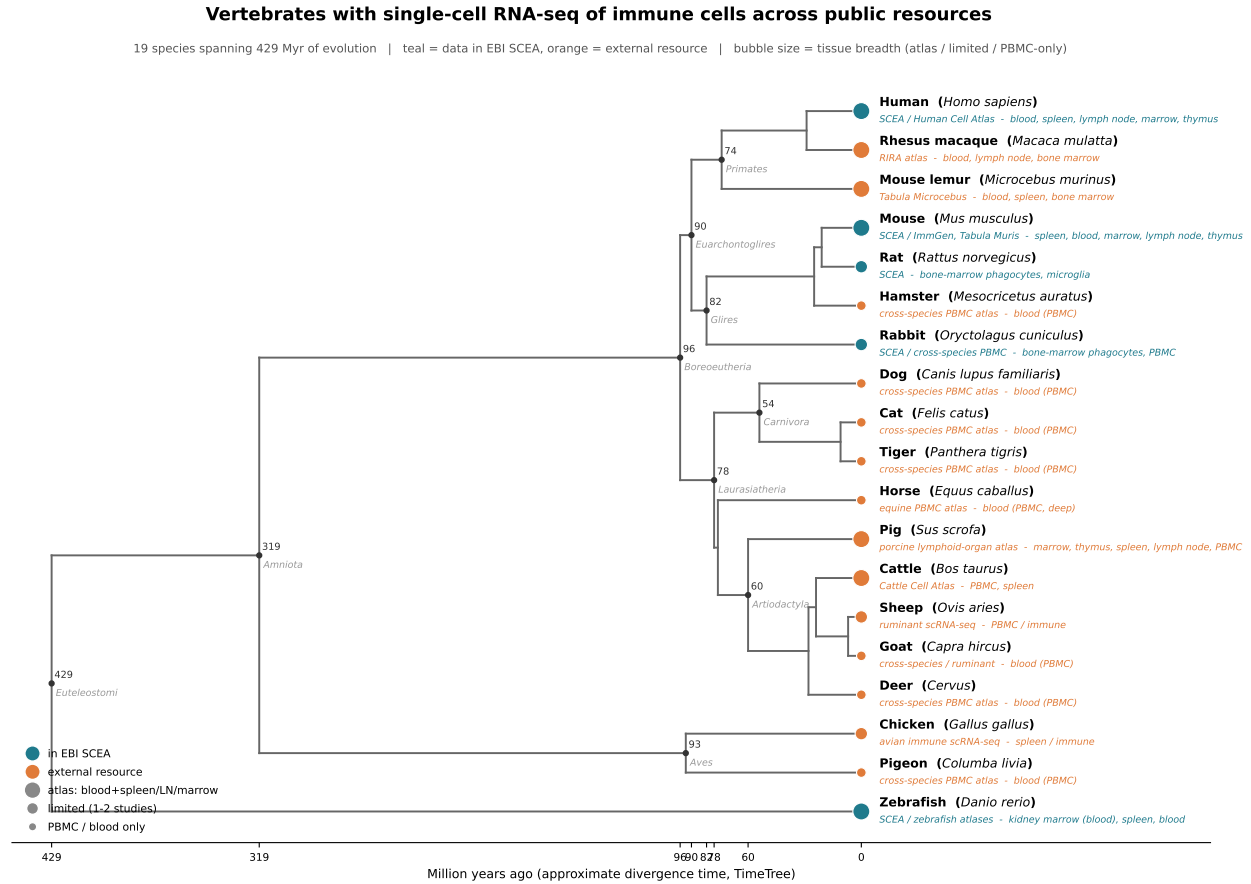


Figure 2: Vertebrates with immune-cell scRNA-seq across *all* public resources. Teal = data in EBI SCEA; orange = external resource. Bubble size encodes tissue breadth: atlas-grade (blood + spleen/lymph node/marrow) > limited (1–2 studies) > PBMC/blood-only. Divergence times are approximate TimeTree values.

Table 2: Vertebrate species with immune-cell scRNA-seq across public resources. “In SCEA” marks species whose data are in the EBI atlas; “coverage” is the tissue breadth (*atlas* = blood + secondary lymphoid + marrow, *limited* = 1–2 immune studies, *pbmc* = blood only). The external set is curated from the literature surveyed and is representative, not exhaustive.

Common	Species	SCEA	Coverage	Resource	Immune tissues
Human	<i>Homo sapiens</i>	✓	atlas	SCEA / Human Cell Atlas	blood, spleen, lymph node, marrow, thymus
Rhesus macaque	<i>Macaca mulatta</i>		atlas	RIRA atlas	blood, lymph node, bone marrow
Mouse lemur	<i>Microcebus murinus</i>		atlas	Tabula Microcebus	blood, spleen, bone marrow
Mouse	<i>Mus musculus</i>	✓	atlas	SCEA / ImmGen, Tabula Muris	spleen, blood, marrow, lymph node, thymus
Rat	<i>Rattus norvegicus</i>	✓	limited	SCEA	bone-marrow phagocytes, microglia
Hamster	<i>Mesocricetus auratus</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Rabbit	<i>Oryctolagus cuniculus</i>	✓	limited	SCEA / cross-species PBMC	bone-marrow phagocytes, PBMC
Pig	<i>Sus scrofa</i>		atlas	porcine lymphoid-organ atlas	marrow, thymus, spleen, lymph node, PBMC
Cattle	<i>Bos taurus</i>		atlas	Cattle Cell Atlas	PBMC, spleen
Sheep	<i>Ovis aries</i>		limited	ruminant scRNA-seq	PBMC / immune
Goat	<i>Capra hircus</i>		pbmc	cross-species / ruminant	blood (PBMC)
Deer	<i>Cervus</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Horse	<i>Equus caballus</i>		pbmc	equine PBMC atlas	blood (PBMC, deep)
Dog	<i>Canis lupus familiaris</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Cat	<i>Felis catus</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Tiger	<i>Panthera tigris</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Chicken	<i>Gallus gallus</i>		limited	avian immune scRNA-seq	spleen / immune
Pigeon	<i>Columba livia</i>		pbmc	cross-species PBMC atlas	blood (PBMC)
Zebrafish	<i>Danio rerio</i>	✓	atlas ⁵	SCEA / zebrafish atlases	kidney marrow (blood), spleen, blood